



EfficientSkinDis: An EfficientNet-based classification model for a large manually curated dataset of 31 skin diseases

Abdul Rafay^a, Waqar Hussain^{b,*}

^a Department of Computer Science, School of Systems and Technology, University of Management and Technology, Lahore, Pakistan

^b Department of Artificial Intelligence, School of Systems and Technology, University of Management and Technology, Lahore, Pakistan

ARTICLE INFO

Keywords:

Deep learning
Skin disease classification
EfficientNet
Medical image analysis
Convolutional neural networks

ABSTRACT

Skin diseases are a common health issue, affecting nearly one-third of the global population, but they are often underestimated in terms of their impact despite being highly visible. Accurately diagnosing skin diseases can be challenging sometimes. Fortunately, deep learning algorithms have shown great potential for a variety of tasks, including the diagnosis of skin diseases. This research presents a novel dataset of 31 skin diseases by blending two different datasets. Three different CNN types, which were EfficientNet, ResNet, and VGG with different architectures, were used for transfer learning on the skin disease dataset. EfficientNet had the highest testing accuracy, thus, it was further trained for fine-tuning. Initially, using the training split of 70%, the EfficientNet model achieved a testing accuracy of 71%. Therefore, as this was considered low, the 70% training split comprising 3,424 samples was further augmented. As a result, the model's accuracy improved to 72%. However, it was observed that the data split of 70/30 was not effective, thus the experiment was re-conducted with a train-test split of 80/20 and an improvement in accuracy score i.e., 74% was observed. To further improve the model's accuracy, augmentations were applied to the data and were able to achieve an accuracy of 87.15%. Finally, the trained model with the best accuracy was deployed on a Streamlit webserver. The proposed model can help society in diagnosing skin disease early, therefore it can be treated timely. The webserver of EfficientSkinDis can be accessed at (<https://abdulrafay97-efficientskindis-app-ooncon.streamlit.app/>).

1. Introduction

Skin weighs around 3.5 kg in the hands of a fully grown adult. It protects the human body against toxins, heat, and UV radiation, among other things [1]. Skin illnesses are far more widespread than we realize. Skin disease affected 84.5 million individuals in America, accounting for one-quarter of the population [2,3]. These skin illnesses are becoming far more dangerous as time passes. According to professionals, if the harm is detected in time, it may be managed. It may get complicated for the dermatologist if he or she uses manual procedures to diagnose disorders. The primary reason for this is that there exist several forms of diseases. Also, many skin illnesses have similar visual features, manual diagnosis can be difficult [4].

Skin is the largest organ in the human body which covers and protects the body. Human skin has multiple functions to perform that include: standardizing your body temperature, fight with bacteria and other viruses to keep them out of the body, helping us feel things, creating vitamin D in response to sunlight, and preventing dehydration

[5]. Skin diseases include disorders that affect the skin. Rashes, inflammation, itching, and other skin abnormalities may result from these disorders [6]. Some skin diseases may be inherited, while others may be caused by environmental factors. Medication, lotions or ointments, or lifestyle changes can be used to treat skin diseases [7].

The symptoms and severity of skin problems vary widely. They might be transient or permanent, and they can be painful or not. Some are caused by environmental factors, while others may be inherited. Some skin disorders are trivial, while others can be fatal. More than 3000 skin illnesses, both acute and chronic, have been reported affecting people of all ages and social backgrounds [8]. Some of them, such as skin cancer, can be fatal. Even if this is not the case, they can impose a significant burden on patients in terms of reduced quality of life and increased expenditures. Skin diseases ranked fourth in terms of non-fatal burden represented as years lost owing to disability [9].

There is a diverse set of illnesses that include common chronic diseases (such as psoriasis and atopic dermatitis) and skin malignancies, as well as unusual ailments such as Geno dermatoses. Because the research

* Corresponding author.

E-mail address: sayhitowaqar@gmail.com (W. Hussain).

<https://doi.org/10.1016/j.bspc.2023.104869>

Received 29 December 2022; Received in revised form 11 March 2023; Accepted 18 March 2023

Available online 11 April 2023

1746-8094/© 2023 Published by Elsevier Ltd.

of uncommon illnesses is dependent on registries, medical literature, or case reports, the technique for investigating the epidemiology of these diverse categories of the population is extremely different [10].

In the past few years, Convolutional Neural Networks (CNNs) have played an important role in object classification and feature learning [11]. The GPU is a very important component to achieving the goals, as the improvement in the architecture of GPUs is rapid these days. The high-performance GPUs have made it possible to map a large dataset on a network for better execution with CNN. A lot of researchers have developed diverse CNN architectures to enhance the performance of CNN output [12]. Also, CNNs have played a significant role in the area of medical image analysis and classification [13–15].

In 2022, Lu et al. aimed to create a computer-based system using artificial intelligence to identify COVID-19 in chest CT images. The authors employed transfer learning to analyze the images and developed a new method called “neighbouring aware representation” to consider the relationships between different parts of the images. The “ILR graph” was then generated by using the k-nearest neighbour algorithm to obtain information about the neighbouring parts of the images and used this information, along with the initial image analysis, to create the new representation. The final system was tested and called “neighbouring aware graph neural network,” using both private and public data sets and found it performed better than existing methods. This system has the potential to be used in clinical settings for detecting COVID-19 [16].

Karthik et al. in 2022, aimed to develop a machine learning model, called Eff2Net, for detecting skin diseases using a Convolution Neural Network (CNN). The model is based on the EfficientNetV2 architecture and uses the Efficient Channel Attention (ECA) block instead of the standard Squeeze and Excitation (SE) block. This results in a reduction in the number of trainable parameters. The model was tested on four types of skin diseases: acne, actinic keratosis (AK), melanoma, and psoriasis, and achieved an overall testing accuracy of 84.70%. The skin is a barrier that protects internal organs and can be damaged by infections or other factors, and a good diagnosis can help with recovery from skin diseases [17].

In 2022, Allugunti found out that melanoma is a type of skin cancer that can be lethal and is more common in Australia and New Zealand than ever before. It is estimated that one in every 15 white New Zealanders will be diagnosed with melanoma at some point in their lives. This research presents a deep learning technique for accurately diagnosing the type of melanoma present in the early stages, which allows for early treatment and can significantly increase survival rates. The proposed model was found to be superior to current state-of-the-art methods in terms of diagnostic accuracy [18].

Hossen et al. in 2022, proposed a study, in which a custom dataset of four types of skin diseases was created and a Convolution Neural Network (CNN) model was proposed and compared with other benchmark CNN algorithms. An image augmentation strategy was also used to increase the size of the dataset and make the model more general. The proposed model achieved high precision and recall rates for acne, eczema, and psoriasis. In addition, a federated learning approach was used to preserve data privacy while maintaining high accuracy in the classification of skin diseases. The combination of CNN-based skin disease classification with the federated learning approach is an effective method for classifying skin diseases while ensuring data security [19].

In 2022, Cai et al. established that diagnosis of skin diseases can be challenging in a clinical setting, and the prevalence of skin diseases is increasing. In this study, a neural network called a multimodal Transformer was proposed for the classification of skin diseases. The multimodal Transformer consists of two encoders for images and clinical metadata and a decoder to fuse the multimodal information. The proposed network uses a Vision Transformer model to extract deep features from images and a Soft Label Encoder to embed metadata, which are treated as labels. A Mutual Attention block in the decoder helps to fuse image and metadata features. The proposed model was evaluated using a private skin disease dataset and the benchmark ISIC 2018 dataset and

was found to perform better than state-of-the-art methods, making it a promising approach for skin disease diagnosis [20].

Anand et al. in 2022 proposed a transfer learning-based model using a pre-trained Xception model, which was modified by adding additional layers. The model was trained and evaluated on the HAM10000 dataset and achieved an accuracy of 96.40% in classifying skin diseases. The proposed model performed particularly well in detecting Benign Keratosis, with a precision of 99%, a sensitivity of 97%, and an F1 score of 0.98. This method can provide patients and doctors with a good understanding of whether or not medical attention is needed [21].

Kshirsagar et al. in 2022, designed a cutting-edge solution that classified skin disorders using LSTM and MobileNetV2. The primary aim of this study was to accurately diagnose skin diseases. Skin problems pose a severe danger to the global fitness network. They grow dangerous when they approach the area being assaulted. Skin disorders are a major concern for the medical team. An alarming number of individuals are suffering skin disorders as air levels increase and diets grow less nutritious. People frequently disregard the warning signs of a skin ailment. The majority of existing methods for diagnosing and treating skin disorders rely on doctor-approved biopsies. A hybrid strategy can help to prevent humans [22].

Lu et al. in 2022 presented a new diagnostic system called CGENet, which uses graph embedding and machine learning to analyze chest CT images and diagnose COVID-19. The system uses a technique called transfer learning to select the best type of analysis to use and incorporates graph theory to further analyze the images. The system is trained using an extreme learning machine and was found to be highly accurate in testing, with an average accuracy of 97.78%. Additionally, the system provides a visual explanation of its diagnostic process through Grad-CAM maps. Overall, the CGENet system is an effective and efficient tool to assist in the diagnosis of COVID-19 [23].

Srinivasu et al. in 2021, reported a study employing deep learning and the MobileNetV2 and LSTM models to automatically diagnose skin disorders proposed. The MobileNetV2 version achieved more accuracy than the LSTM. The suggested approach performed well in terms of retaining state data for particular projects. The grey degree co-excitability grid was used to measure the development of sickness increase. Convolutional neural networks (CNN), Fine Tuned Neural Networks (FTNN), and Visual Geometry Group’s Deep Geometric Identification for Huge Image Recognition all performed similarly (VGG) [24].

In 2021, Lu et al. presented a new method for detecting abnormal brain activity using magnetic resonance images was developed. The method starts by modifying a pre-trained computer model known as AlexNet and training it using brain images. Then, the final layers of the model are replaced with a machine learning technique called extreme learning machine. A special algorithm was created to determine the optimal number of layers to replace. Finally, the extreme learning machine is further optimized using a chaotic bat algorithm to improve its accuracy. Testing the method using a specific validation technique, known as 5x hold-out validation, showed that it performed better than other existing methods [25].

Voggu and Rao in 2020, proposed a study in which three distinct skin disorders, comprising herpes, dermatitis, and psoriasis, were identified using a unique identification technique. Skin snapshots are first pre-processed with filtering and modification to remove noise and unwanted inheritance. Deep neural community-based techniques have been used to autonomously diagnose skin disorders using dermoscopic images. This was necessary to develop automated ways for improving the accuracy of analysis for diverse skin types and psoriasis conditions [26].

In 2019, Burlina et al. proposed a method that used cross-sectional picture datasets to train an intensive version to perform an erythema migrants’ class against various skin ailments such as Tania Corpus and Shingles, and also non-pathogenic skin diseases. The machinability to classify skin types was also evaluated using verification photographs. It



Fig. 1. Workflow of the proposed study.

is also being suggested to employ the erythema migrant detection machine in non-medical human representations. An ongoing large study of people with acute Lyme disease who were enrolled in 2016 and 2017 gathered and combined publicly accessible online pictures of erythema migrants and Lyme perplexing skin lesions with photos of erythema migrants [27].

Monisha et al. in 2019 reported a study based on the neuronal community which examined the use of several similarities in skin texture and symptoms, including measles (rubella), German measles (rubella), and chickenpox. Symptoms of some skin conditions may be similar. Measles (rubella), chickenpox, and German measles, for example, all look like skin rashes. Skin diseases are difficult to diagnose since the victim's past medical histories, physical examination records, and relevant laboratory diagnostic methods must all be considered. In such cases, computer-assisted diagnostic and identification approaches may be available. The laptop rolls comprise image processing, image extraction of features, and fact classification using the Artificial Neural Network (ANN) category [28].

Bhavani et al. in 2019, proposed a way for diagnosing different skin disorders. Mobile net, Inception V3, and ResNet are three examples. To study and test medical photographs, a set of machine learning concepts, particularly logistic regression, was used, and deep learning algorithms were used to extract attributes from a scientific image. Excellent performance may be obtained by combining three convolutional neural networks in a hybrid design [29].

ALenezi in 2019, presented an image-processing-based technique for diagnosing skin diseases. The procedure entailed building a virtual image of the affected skin area and then analyzing photographs to determine

the type of sickness. The proposed method became easier and faster, and it no longer required any expensive equipment other than a camera and a laptop. The shadow image is used as input for this procedure. The image is then scaled for a pre-qualified advanced neural network to extract information. This function was termed Multi Elegance SVM later on. Finally, the individual is assessed based on the consequences, kind of disease, progress, and severity [30].

Velasco et al. in 2019, proposed a model for skin diseases whereas the mobile net accuracy was improved by employing unique sampling methods and pre-processing of input data. Using basic sample procedures and the raw data's default pre-processing, it was 84.28% accurate. Even with skewed datasets and conventional input record pre-processing, accuracy was 93.6%. The model's accuracy increased to 91.8% once oversampling was discovered in the dataset [31].

Hameed et al. in 2018, suggested a smart diagnosis strategy for a more appealing class of skin and prickly lesions. The proposed solution was implemented using hybrid methods, specifically an error-correcting deep convolution neural network and outcome codes (ECOC) based on a useful support vector machine (SVM). The technique provided was complicated, requiring skin lesions to be classified into one of 5 categories: benign, benign, or malignant. The study made use of 9,144 pictures gathered from common assets. The feature was extracted using Alex NET, a CNN-approved technique [32].

The present study proposes a novel method for the identification of multiple skin diseases using CNNs to provide an efficient and accurate prediction model of skin diseases, instead of using traditional laborious and expensive experimental tools. The dataset used in this study is curated from different sources, comprising 31 diseases, and is collected

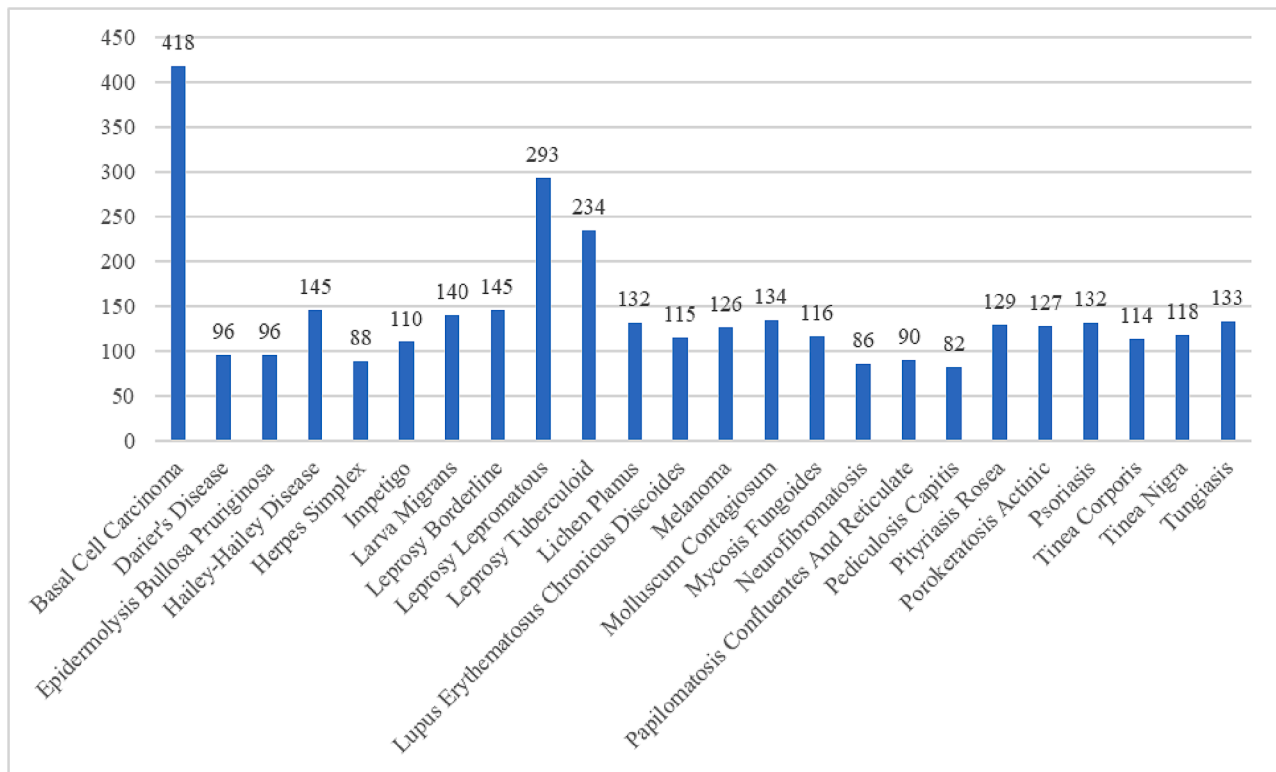


Fig. 2. Dataset distribution of 24 classes.

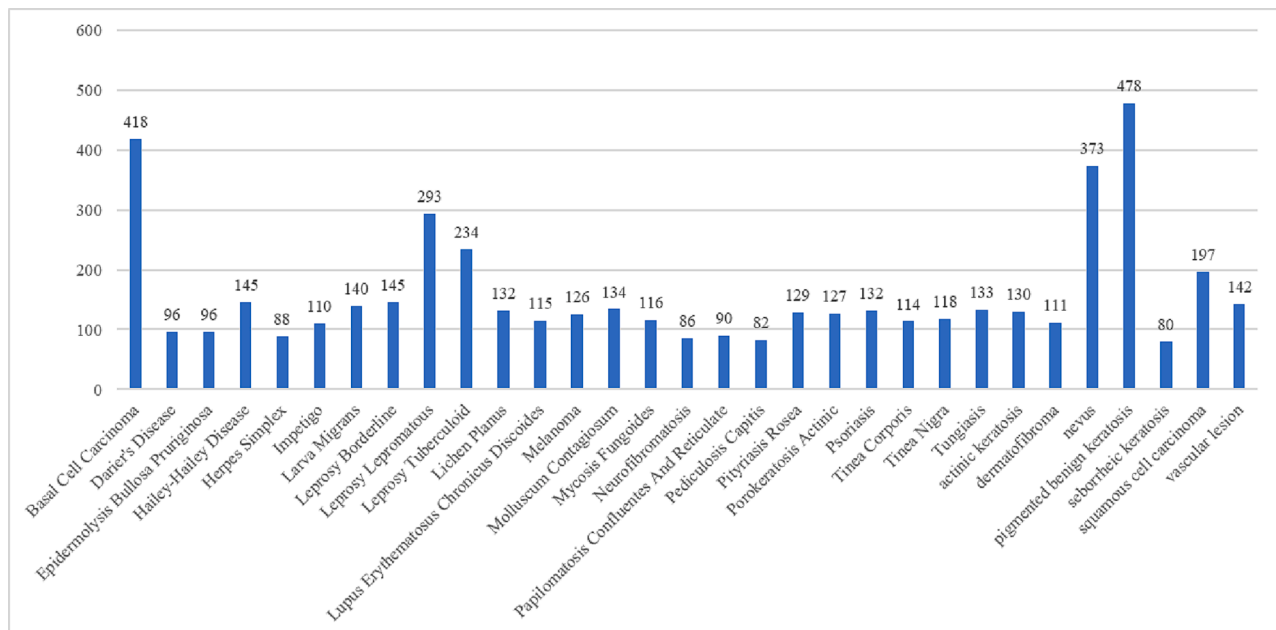


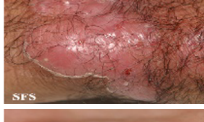
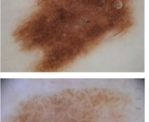
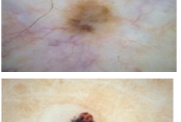
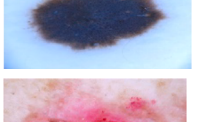
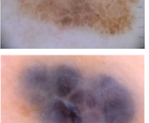
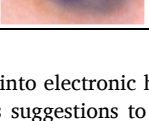
Fig. 3. Dataset Distribution of 31 classes.

keeping in mind the quality of the dataset. It is evident in the literature that no previous study has been reported in which a large number of skin diseases are classified due to the complexity and challenging nature of the task. Furthermore, to evaluate the potential of the model. Exhaustive performance evaluation was performed and the top-performing model was deployed on a webserver for making it publicly available.

The unique aspect of the proposed study lies in the fact that no previous research has been conducted using such a large dataset within this domain. The practical significance of this proposed study is of great

value to healthcare organizations. The high accuracy of the model in classifying skin diseases can assist these organizations in enhancing their diagnostic abilities and delivering more efficient treatments to patients. This can lead to improved patient outcomes and decreased healthcare expenses. Health organizations can use the findings of this research to develop a mobile application for dermatologists to diagnose skin diseases quickly and accurately. This could save time and resources, as well as it will increase patients' satisfaction, as they would receive a diagnosis and treatment plan faster. Another example could be the

Table 1
Dataset Samples and Respective Labels.

Basal Cell Carcinoma		Leprosy Lepromatous		Papilomatoss Confluentes And Reticulate	
Darriers		Leprosy Tuberculoid		Pediculosis Capitis	
Epidermolysis Bullosa Pruriginosa		Lichen Planus		Pityriasis Rosea	
Hailey-Hailey Disease		Lupus Erythematosus Chronicus Discoides		Porokeratosis Actinic	
Herpes Simplex		Melanoma		Psoriasis	
Impetigo		Molluscum Contagiosum		Tinea Corporis	
Larva Migrans		Mycosis Fungoides		Tinea Nigra	
Leprosy Borderline		Neurofibromatosis		Tungiasis	
Actinic Keratosis		Dermatofibroma		Nevus	
Pigmented Benign Keratosis		Seborrheic Keratosis		Squamous Cell Carcinoma	
Vascular Lesion					

integration of this model into electronic health records (EHRs) to provide automated diagnosis suggestions to healthcare providers, which would reduce the likelihood of misdiagnosis and improve patient outcomes. There is a growing demand for automated skin disease diagnosis, which can help reduce the workload of dermatologists and improve the speed and accuracy of diagnoses. The rest of the paper is organized into 3 sections. [Section 2](#) contains details of materials and methods; [Section 3](#) describes results and discussion while [Section 4](#) concludes the study.

2. Materials and methods

[Fig. 1](#) represents the workflow of this proposed study. To get started with the research, domain analysis was done on the existing literature.

Mainly it was focused on identifying the gaps and approaches of previous studies. The dataset was curated from two sources which were Atlas Dermatology and ISIC. The dataset was further split with 2 different splitting ratios which were 70/30 and 80/20. After splitting the training and testing dataset, the training data of each split was augmented using 6 types of augmentations. Augmentations were used to up-sample the training dataset and after augmentations, each training dataset was trained on 12 different models belonging to 3 CNN architectures. The top model was validated using 6 different measures as mentioned in the flowchart. The top-scoring model was then deployed on Streamlit for people to use.

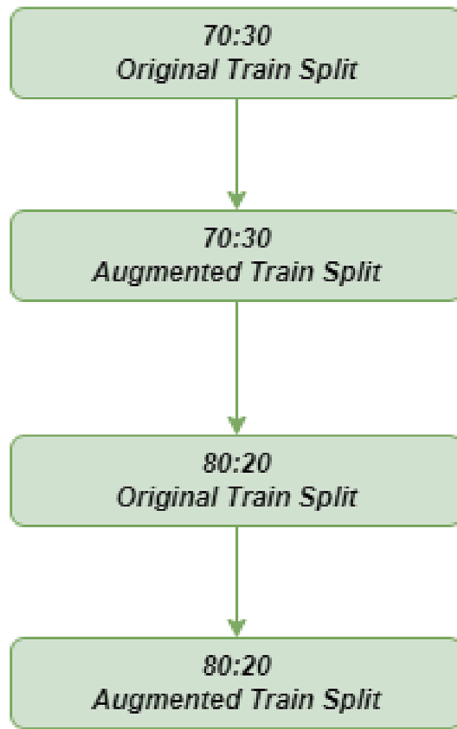


Fig. 4. Data splits used in this study.

2.1. Dataset collection

The dataset is known to be an important component to train any intelligent model. The deep learning model requires a lot of data to train the best model. The dataset should be rich and standardized to get the good performance of a Deep Learning model. The quality of data is equally important as the quantity of the dataset. Even if one has put together a reasonably good algorithm but the data is not enough or rich, the model's training accuracy would not be that good.

Keeping these factors in mind, the dataset was manually curated from two major resources which were Atlas Dermatology (<https://www.atlasdermatologico.com.br/>) and ISIC (<https://www.isic-archive.com/>). Atlas dermatology contained 561 skin diseases in total, while some of the skin diseases contained a quite low number of samples, insufficient for training and learning for a Deep Learning model. Even, some of the classes contained only 9 to 10 samples. Therefore, a threshold was set and to manually collect and curate the dataset, a custom web scraper was used which collected the data of the classes containing at least 80 samples. After the filtration process up to a threshold, the dataset contained 24 classes whereas these 24 classes contained 3,399 samples in the whole dataset. The overall distribution of data in 24 classes is shown in Fig. 2.

The second source i.e., ISIC, contained 9 classes of skin diseases. However, out of these 9, 2 classes were already present in the Atlas Dermatology dataset. Therefore, those 2 classes were excluded from the 9 classes before merging. After this filtration process, the dataset

Table 2

Details of the number of samples for all splits.

Splitting and Augmentation Scheme	Train Samples	Test Samples
70:30	3,424	1,486
70:30 (with Training Data Augmentation)	33,483	1,486
80:20	3,916	994
80:20 (with Training Data Augmentation)	44,918	994

Table 3

Details of the models and their hyper-parameters.

Models	Learning Rate	Weight Decay	Models	Learning Rate	Weight Decay
EfficientNet-B0	5e-4	1e-6	EfficientNet-B6	4e-4	1e-6
EfficientNet-B1	5e-4	1e-6	ResNet-50	3e-4	1e-6
EfficientNet-B2	5e-4	1e-6	ResNet-101	5e-4	1e-6
EfficientNet-B3	3e-4	1e-6	ResNet-152	2e-4	1e-6
EfficientNet-B4	2e-4	1e-6	VGG-16	5e-4	1e-6
EfficientNet-B5	3e-4	1e-6	VGG-19	5e-4	1e-6

Table 4

Results of all the trained models on 70:30 split (Sorted based on Performance).

Model	Train (Accuracy)	Test (Accuracy)	Epoch
EfficientNet-B2	0.9804	0.7144	15
EfficientNet-B3	0.9815	0.7128	15
EfficientNet-B4	0.9837	0.7119	15
EfficientNet-B0	0.9704	0.7108	15
EfficientNet-B6	0.9867	0.7104	15
EfficientNet-B1	0.9634	0.7060	15
EfficientNet-B5	0.9706	0.7004	15
ResNet-101	0.9012	0.5260	10
ResNet-50	0.9314	0.5159	10
VGG-16	0.8645	0.5129	20
ResNet-152	0.8821	0.5062	20
VGG-19	0.9131	0.4658	20

Table 5

Results of all the trained models on 70:30 split with training dataset augmentation (Sorted based on Performance).

Model	Train (Accuracy)	Test (Accuracy)	Epoch
EfficientNet-B2	0.9767	0.7241	15
EfficientNet-B6	0.9837	0.7212	15
EfficientNet-B3	0.9825	0.7177	15
EfficientNet-B4	0.9884	0.7114	15
EfficientNet-B0	0.9804	0.7108	15
EfficientNet-B1	0.9814	0.7060	15
EfficientNet-B5	0.9816	0.7004	15
ResNet-101	0.8962	0.5260	10
ResNet-50	0.9256	0.5159	10
ResNet-152	0.8825	0.5062	20
VGG-19	0.9130	0.4658	20
VGG-16	0.8693	0.4129	20

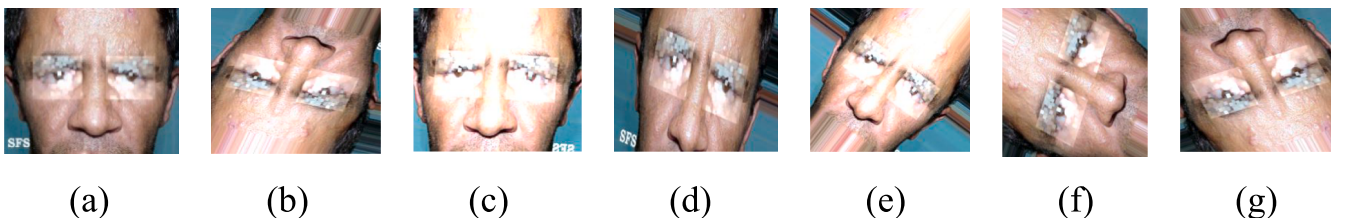


Fig. 5. Samples of Augmented Data where (a) Original (b) Rotation (c) Shear (d) Center Zoom (e) Brightness (f) Horizontal Flip and (g) Vertical Flip.

Table 6

Results of all the trained models on an 80:20 split (Sorted based on Performance).

Model	Train (Accuracy)	Test (Accuracy)	Epoch
EfficientNet-B2	0.9826	0.7436	15
EfficientNet-B0	0.9877	0.7434	15
EfficientNet-B3	0.9815	0.7375	15
EfficientNet-B1	0.9872	0.7310	15
EfficientNet-B4	0.9915	0.7237	15
EfficientNet-B6	0.9861	0.7219	15
EfficientNet-B5	0.9863	0.7217	15
ResNet-50	0.9964	0.6995	10
ResNet-101	0.9947	0.6940	10
ResNet-152	0.9665	0.6916	20
Vgg-16	0.9449	0.6362	20
Vgg-19	0.9453	0.6325	20

Table 7

Results of all the trained models on 80:20 split with training dataset augmentation (Sorted based on Performance).

Model	Train (Accuracy)	Test (Accuracy)	Epoch
EfficientNet-B2	0.9861	0.8715	15
EfficientNet-B4	0.9873	0.8445	15
EfficientNet-B3	0.9825	0.8230	15
EfficientNet-B1	0.9839	0.8180	15
EfficientNet-B0	0.9815	0.8167	15
EfficientNet-B5	0.9834	0.8167	15
EfficientNet-B6	0.9832	0.8089	15
ResNet-101	0.9967	0.7750	10
ResNet-50	0.9924	0.7591	10
ResNet-152	0.9615	0.7213	20
VGG-16	0.9489	0.5762	20
VGG-19	0.9436	0.4325	20

contained 1,511 samples in 7 classes. The dataset collected from both sources was merged into a single dataset and the final dataset contained 31 classes, comprising 4,910 samples in the whole dataset. The distribution of data in 31 classes is shown in Fig. 3. A sample for each class is shown in Table 1. However, the number of samples was still very low collectively, thus augmenting the dataset.

2.2. Dataset Pre-Processing and augmentations

As mentioned earlier, the dataset had collectively low samples, thus, to address this issue, augmentation was performed on images. However, before that, the dataset was divided into the train-test-split so that the test dataset did not contain any augmented samples. Fig. 4 represents the mechanism of splitting the dataset in various splits.

In this study, two different splits were done which were 70:30 and 80:20. However, in terms of 70:30, this study did not achieve good results later in experimentations, thus, finalized the model with an 80:20 split. Here, 80% of the data was considered training data, while 20% was

considered testing data. 80% of the data had 3,916 data from all 31 classes, whereas testing data had 994 samples. Thus, to increase samples in the training split, the dataset was augmented to have sufficient samples. A total of 6 augmentations, which were Center Zoom, Rotation, Brightness, Shear, Vertical Flip, and Horizontal Flip, were performed for the training dataset. Samples of augmentations are shown in Fig. 5. After the augmentation process, the training dataset consisted of 37,551 samples of skin disease images for 31 classes. After all the evaluations, as reported in Section 3, it was observed that the 80:20 split gave the highest score and it was considered the final dataset. The number of samples for all splitting approaches is shown in Table 2.

The final 80:20 split of the dataset is available on Google Drive (https://drive.google.com/drive/folders/1AiDVpgy-o4ZLKXZ_yqnWWEH_FYhCbrfP1).

2.3. Training convolutional neural networks

For the present study, a total of 12 CNNs were trained on different splits of the dataset that was collected in the previous steps. Each model was trained to 10 epochs to evaluate which model performed well. Learning rate and weight decay were changed depending on the architecture of the model (Table 3).

EfficientNet-B2 was the best-scoring model on 10 epochs. Therefore, EfficientNet-B2 was further trained to its converging point where accuracies were not changing. The extensive training took the previous testing accuracy of 10 epochs from 71.3% to 87.15%.

2.4. Evaluation metrics method

Initially, the confusion matrix was computed to get the scores of TP, TN, FP and FN for all classes. As the proposed study comprised the multi-class dataset, all four measures were computed as one versus other classes. In all cases, if the score is computed for any of the classes, the remaining 30 classes out of 31 acted as negative classes. TP represents a correct and positive prediction for any class made by the model. FP represents an incorrect and positive prediction. FN represents an incorrect and negative prediction, whereas TN represents a correct and negative prediction. These terms are used in combination to calculate different performance metrics, such as accuracy, precision, recall, and F1 score, that give a comprehensive evaluation of the model's performance.

For performance evaluation, well-discussed and previously reported performance evaluation metrics including accuracy precision, recall, F1-score, and support were used. Precision is the classification model's capacity to detect only relevant data elements. To calculate it, the total number of true positives is divided by the overall number of true positives + false positives, as shown in Eq. (1).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

The capacity of a classification model to find all relevant instances in

Table 8

Results of all evaluation metrics for the trained models on the respective dataset (Sorted based on Performance).

Model	Train (Accuracy)	Test (Accuracy)	Epoch	Precision	Recall	F1-score	Support
EfficientNet-B2	0.9861	0.8715	15	0.87	0.87	0.87	994
EfficientNet-B4	0.9873	0.8445	15	0.84	0.84	0.84	994
EfficientNet-B3	0.9825	0.8230	15	0.82	0.82	0.82	994
EfficientNet-B1	0.9839	0.8180	15	0.82	0.82	0.82	994
EfficientNet-B0	0.9815	0.8167	15	0.82	0.82	0.82	994
EfficientNet-B5	0.9834	0.8167	15	0.82	0.82	0.82	994
EfficientNet-B6	0.9832	0.8089	15	0.81	0.80	0.80	994
ResNet-101	0.9967	0.7750	10	0.78	0.77	0.77	994
ResNet-50	0.9924	0.7591	10	0.76	0.7	0.76	994
ResNet-152	0.9615	0.7213	20	0.73	0.72	0.72	994
VGG-16	0.9489	0.5762	20	0.58	0.58	0.58	994
VGG-19	0.9436	0.4325	20	0.45	0.43	0.43	994

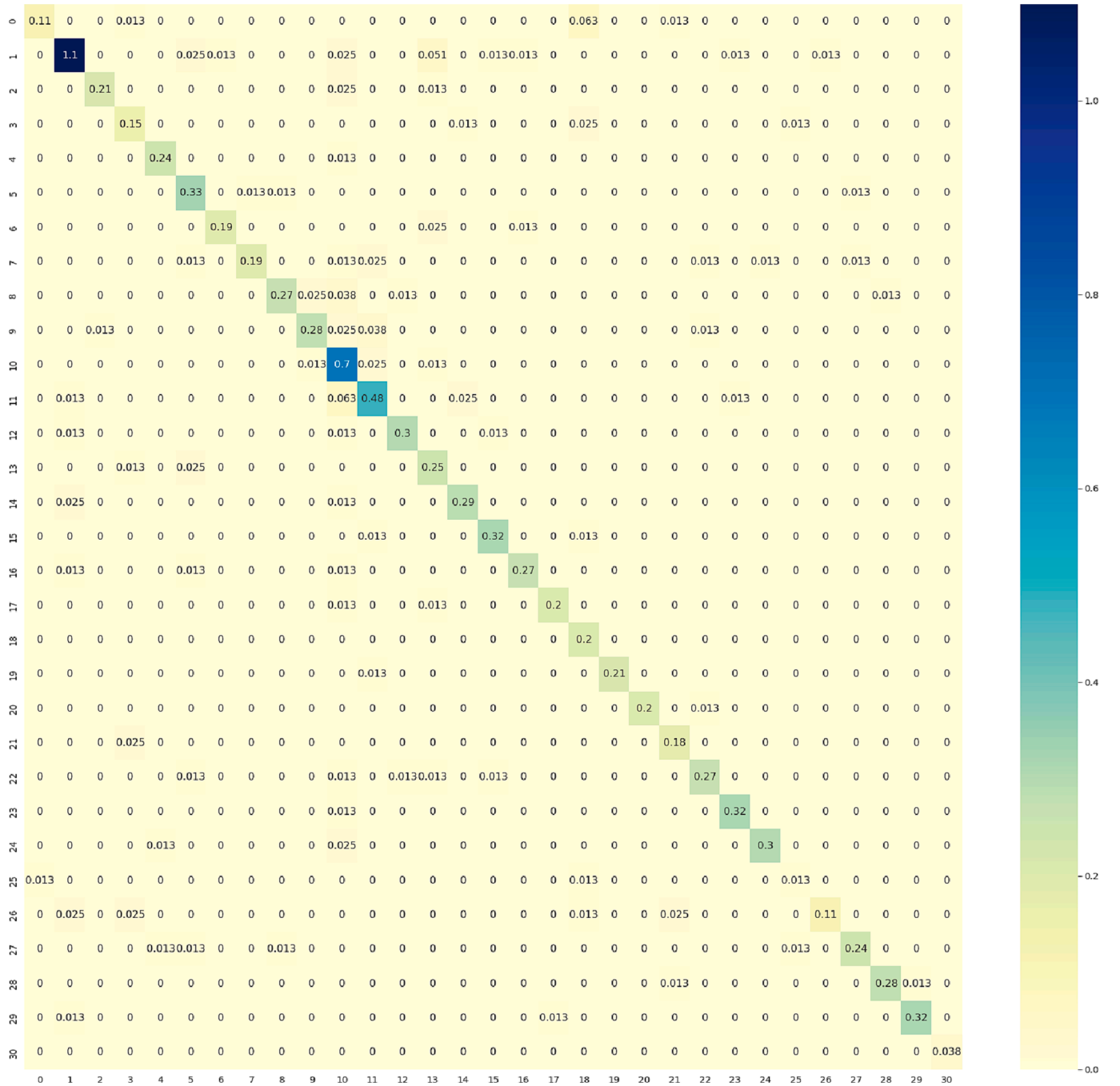


Fig. 6. Confusion Matrix for EfficientNet-B2.

a dataset is known as recall. To determine recall, divide the number of true positives by the total number of true positives and false negatives, as shown in Eq. (2).

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

The F1 score is the mean of precision and recall when both metrics are used and is shown in Eq. (3).

$$F1 - Score = \frac{2 * (Precision * Recall)}{Precision + recall} \quad (3)$$

Whereas, the support is defined as the number of actual instances of the class in the provided dataset.

2.5. Deployment

The model was made available as a webserver through Streamlit. Streamlit is a Python library that enables developers to create web applications and data visualizations using simple Python scripts. The model's weights were saved in an H5 file after the training. All the relevant files were uploaded on GitHub (<https://github.com/abdulrafay97/EfficientSkinDis>) to integrate the webserver on Streamlit. The webserver was designed to take a picture as input, which was further converted to a PIL image object and then return a response of the predicted label after passing through the PIL image object from the classifier.

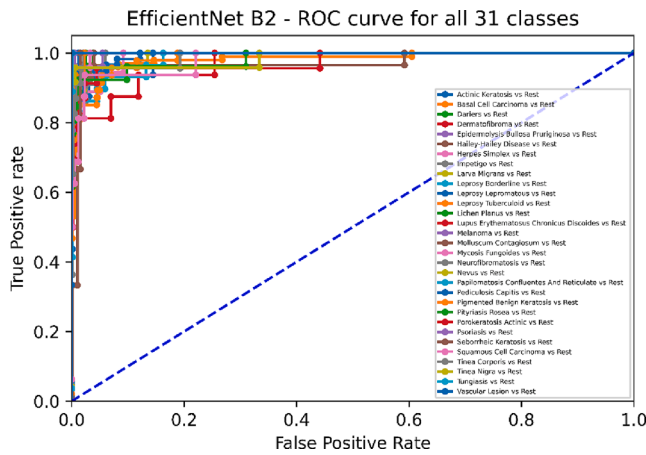


Fig. 7. ROC Curves for EfficientNet-B2.

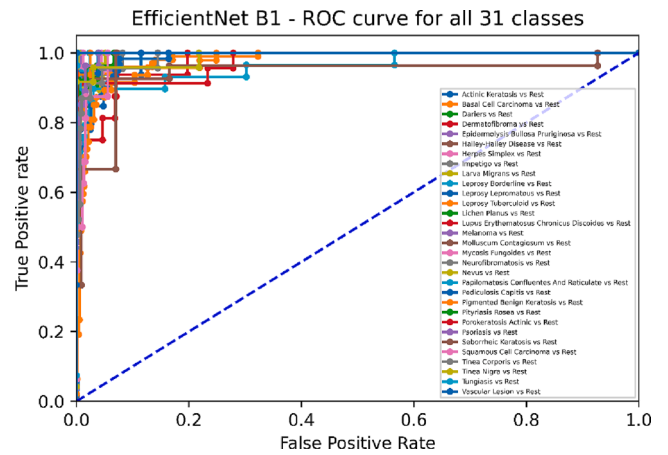


Fig. 10. ROC Curves for EfficientNet-B1.

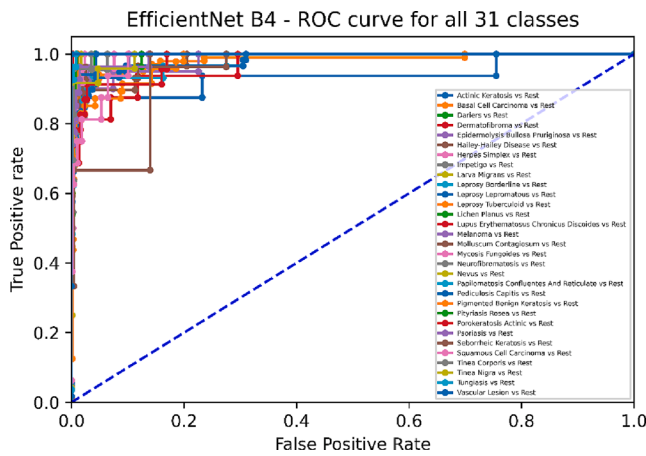


Fig. 8. ROC Curves for EfficientNet-B4.

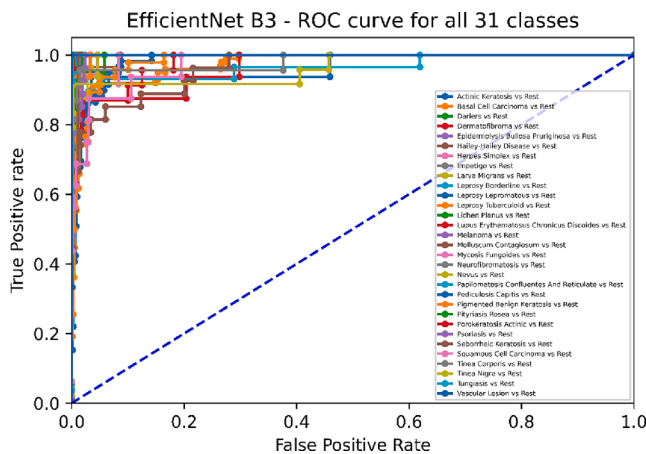


Fig. 9. ROC Curves for EfficientNet-B3.

3. Results and discussion

In this study, transfer learning was applied to train 12 different CNNs on the 31 skin diseases. Out of all these architectures, EfficientNet achieved the best testing accuracy.

3.1. Experimental setup

For this study, NVIDIA GeForce RTX 3080 was utilized for the experiment. A virtual environment was created with Python 3.6, PyTorch, and CUDA 1.6 and other required Python libraries were added. The NVIDIA GeForce RTX 3080 has a memory capacity of 10 GB. For implementation, PyTorch was used which is a Deep Learning tensor library written in Python that is typically used in CPU and GPU operations. PyTorch is recommended above other deep learning frameworks such as Tensor Flow and Keras because it uses dynamic computation graphs and therefore is purely Pythonic.

3.2. Evaluation of different data splits

The initial training was performed on the 70:30 split of the dataset using as the default for training, the highest accuracy attained was 71.44% for EfficientNet-B2 (Table 4). Therefore, to improve the scores, 70% of the training dataset was further augmented, but in this case, EfficientNet-B2 managed to achieve a testing accuracy of (72.41%) which was not up to the benchmark scores previously reported (Table 5).

To overcome this, the dataset was partitioned to 80:20. The modification in the dataset split resulted in a significant increase in testing accuracy for EfficientNet-B2 which was now 74% (Table 6). As a consequence, 80% of the training data was further augmented to attain better outcomes. The EfficientNet-B2model achieved an accuracy of 87.15% after the augmentations. The results are reported in Table 7. It is pertinent to mention here that in all cases, EfficientNet-B2 was the top-scoring model. EfficientNet-B2 uses compound scaling to adjust its capacity automatically. This allows it to adapt to various input sizes and computational constraints without needing manual tuning or extensive hyperparameter search. For further evaluation, we used the 80:20 split with augmentations as the final dataset.

3.3. Evaluation of the models

For further performance evaluation, only the final dataset with an 80:20 split as the top-scoring model was used for training and computing the evaluation metrics scores for all models. The complete results of the evaluation are shown in Table 8.

Table 8 displays the evaluation findings. A total of 12 models from three different architectures were trained. Every model was trained to its converging point where the accuracies no longer increased or decreased. The best-performing model was EfficientNet-B2. As it managed to contain a very good testing accuracy. EfficientNet-B2 is a type of neural network developed by Google researchers that is part of the Efficient Nets family. These models aim to improve upon existing convolutional

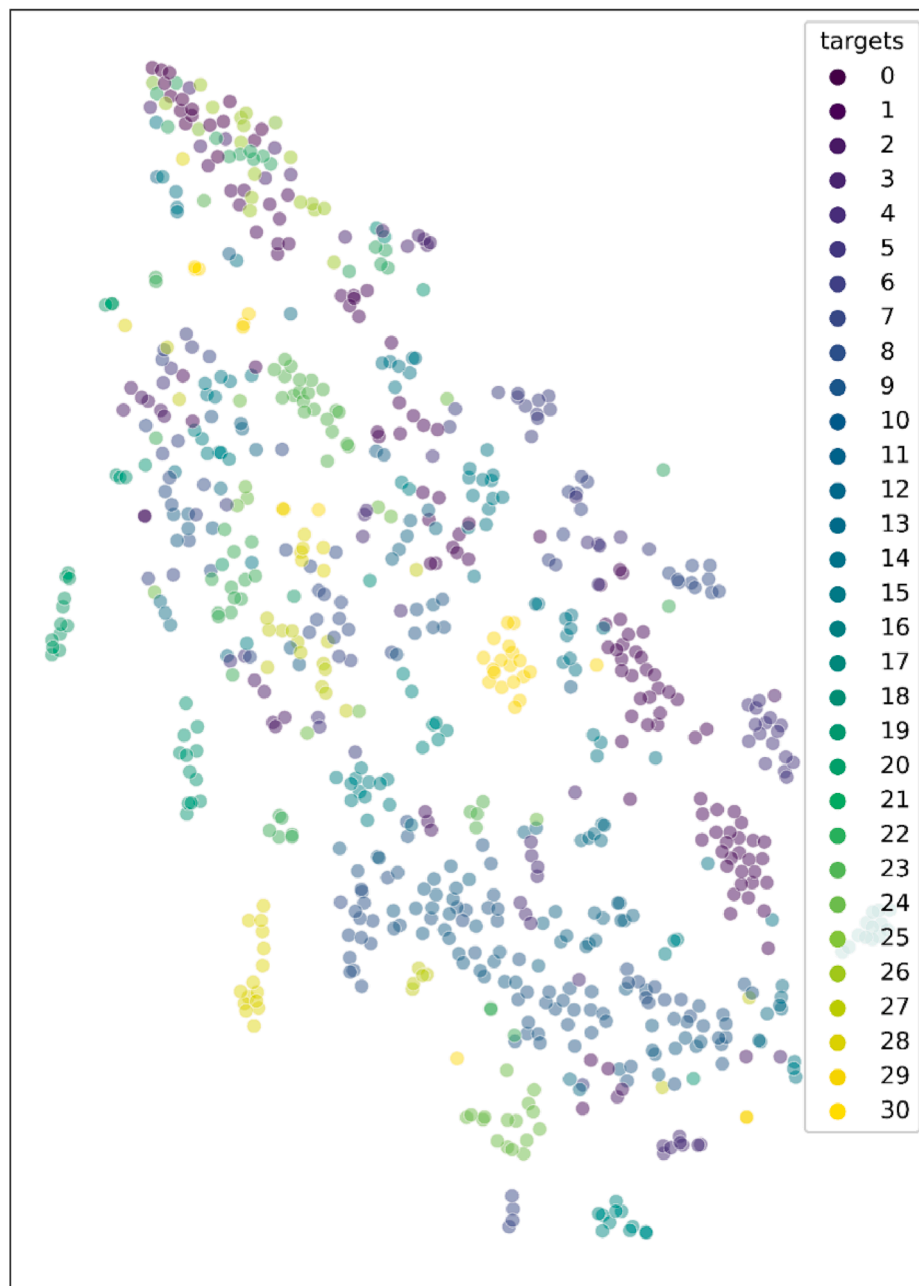


Fig. 11. T-SNE Plot for EfficientNet-B2.

neural networks by using fewer computational resources and fewer parameters.

EfficientNet-B2's design incorporates two key ideas: depth-wise separable convolutions and compound scaling [33]. Depth-wise separable convolutions split the convolutional operation into two stages, allowing the network to be more efficient. Compound scaling uses predefined scaling coefficients to automatically adjust the network's capacity to the input size and available computation. This eliminates the need for manual tuning or extensive hyperparameter search. EfficientNet-B2 uses compound scaling to adjust its capacity automatically. This allows it to adapt to various input sizes and computational constraints without needing manual tuning or extensive hyperparameter search [34].

The confusion matrix provided in Fig. 6 demonstrates, which diseases the model was able to identify with the highest accuracy. It shows how effective the model is by providing a visual representation of the percentage of correct predictions made by the best-trained model for

each disease. The diagonal elements of the matrix represent the proportion of correctly identified cases.

In Fig. 7, Fig. 8, Fig. 9, and Fig. 10, ROC curves are provided for 4 top performing models which were EfficientNet-B2, EfficientNet-B4, EfficientNet-B3 and EfficientNet-B1. Here, the curves provided are for all 31 classes while the mean area under the curve for all ROC curves for identifying 31 classes of each model were 0.91, 0.90, 0.89 and 0.87, respectively.

T-SNE is a method for translating high-dimensional data into a lower-dimensional domain in order to simplify it. This is accomplished by searching for patterns in the data and clustering similar points together while isolating different ones. This approach is extremely effective for viewing large amounts of data. Fig. 11 shows the T-SNE plot graph for the best-performing model (EfficientNet-B2). It is evident from the T-SNE plot that the model learnt distinguishable representations for all classes to create a boundary of classification for all classes.

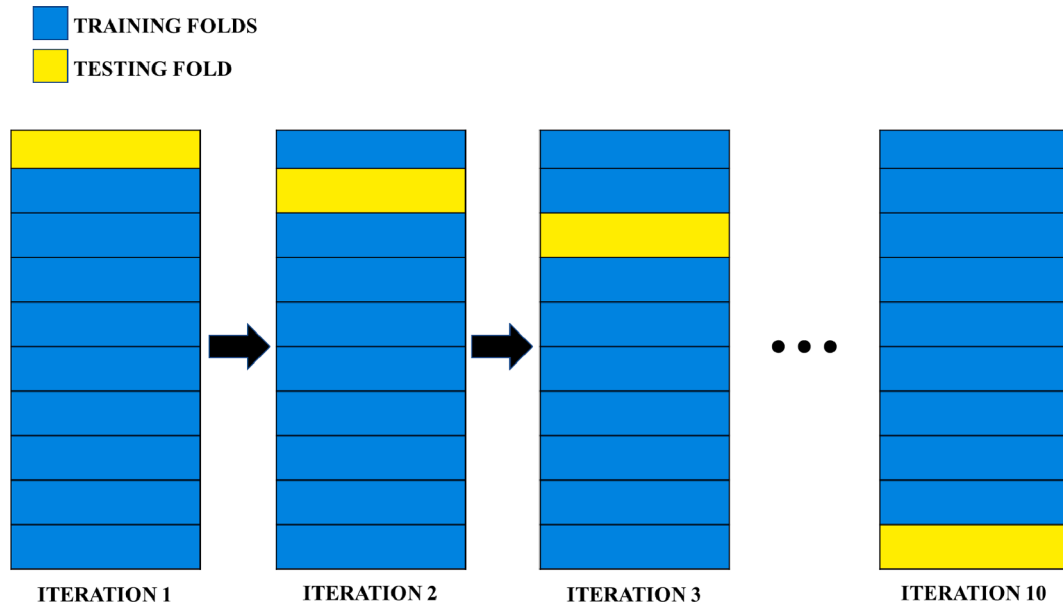


Fig. 12. Graphical Depiction of 10-fold Cross Validation.

Table 9

Results of 10-fold Cross-Validation for EfficientSkinDis.

Iteration of 10-Fold	Test Accuracy	Precision	Recall	F1-score
Iteration-1	0.8715	0.87	0.87	0.87
Iteration -2	0.8518	0.86	0.85	0.86
Iteration -3	0.8627	0.85	0.85	0.87
Iteration -4	0.8511	0.86	0.86	0.85
Iteration -5	0.8613	0.86	0.85	0.86
Iteration -6	0.8543	0.85	0.85	0.87
Iteration -7	0.8625	0.85	0.85	0.85
Iteration -8	0.8702	0.87	0.85	0.87
Iteration -9	0.8687	0.85	0.84	0.85
Iteration -10	0.8659	0.86	0.85	0.85

3.4. Validation of EfficientNet-B2 through 10-fold Cross-Validation

To further validate the proposed model of EfficientSkinDis, the protocol of 10-fold cross-validation was employed. In 10-fold cross-

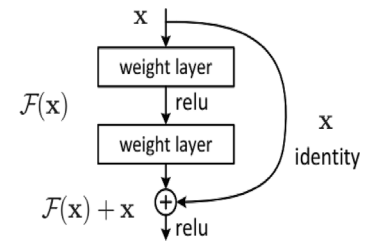


Fig. 14. Skip connections visual representation of ResNet [39].

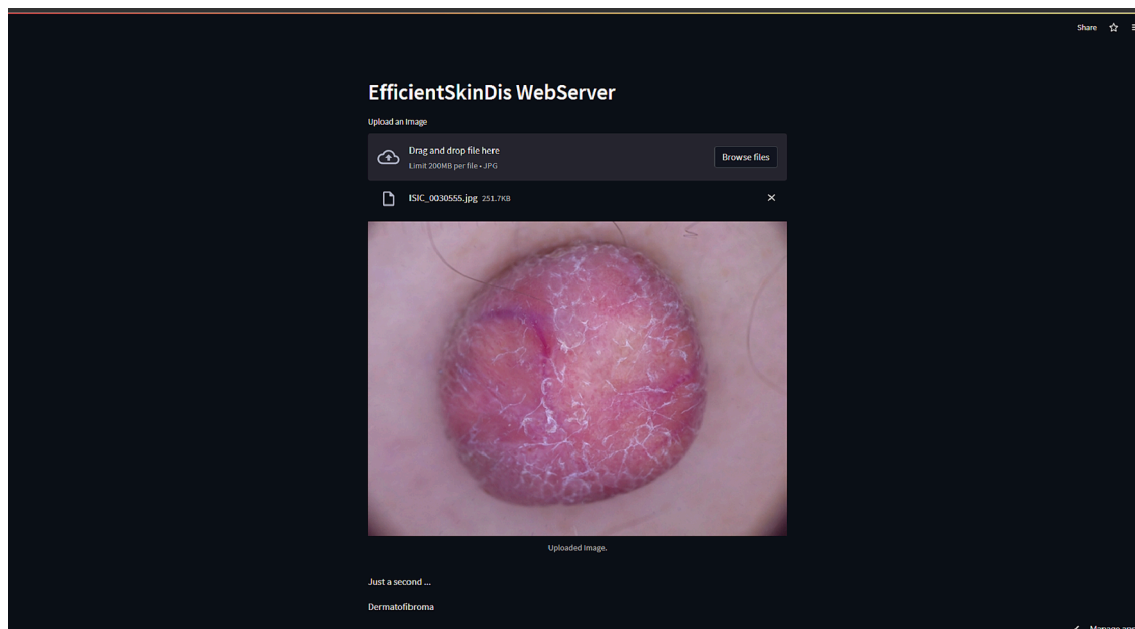


Fig. 13. Web server interface for EfficientSkinDis.

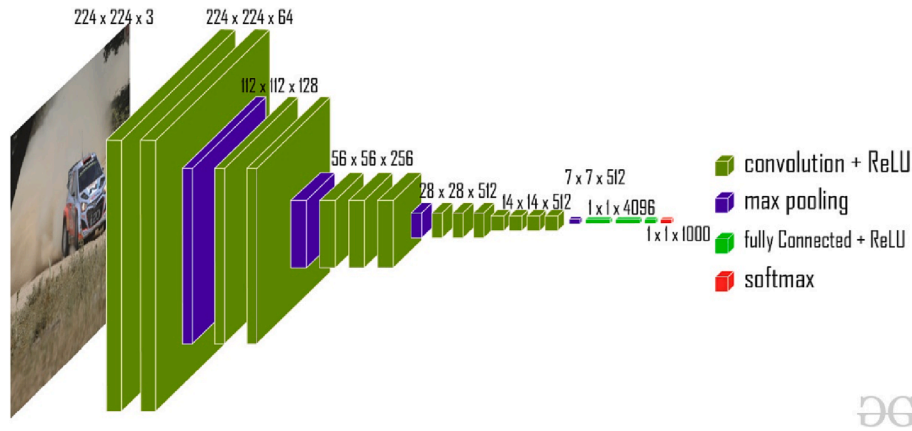


Fig. 15. VGG Architecture [40].

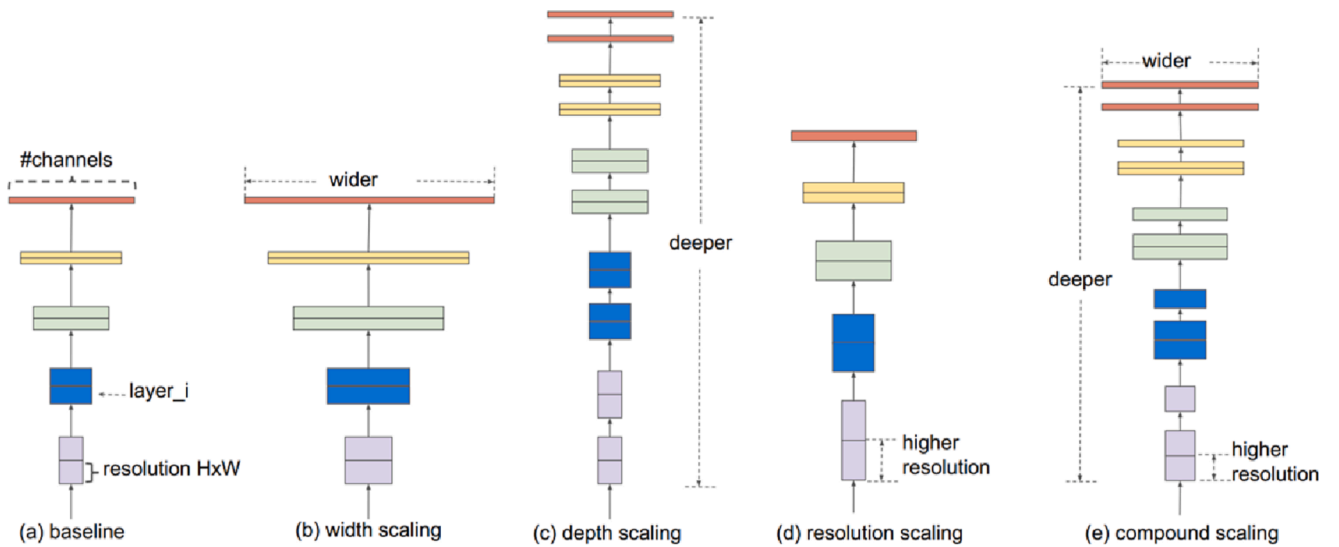


Fig. 16. EfficientNet Architecture [33].

validation, the whole dataset is provided as input, and it is split into 10 disjoint folds. The training and testing are performed 10 times, and in each iteration, 9 folds are used for training while the remaining 1 fold is used for testing. However, in each iteration, the training folds and 1 test fold keep changing. Thus, for all 10 iterations, all 10 folds are used for testing, iteratively. For better understanding, a graphical depiction of 10-fold cross-validation is provided in Fig. 12.

For EfficientSkinDis, the original dataset of 31 classes, comprising 4,910 samples (without training dataset augmentations) was employed. The dataset was initially used without augmentation because it would have not been a fair evaluation if augmented samples were split into the testing fold. The dataset was split into 10 disjoint folds comprising 491 samples each. For all 10 iterations, a total of $491 \times 9 = 4419$ samples were used for training while 491 samples of the remaining 1 fold were used for testing. However, to improve the performance of the model, the training dataset of 4419 samples was augmented with 6 augmentations, which were Center Zoom, Rotation, Brightness, Shear, Vertical Flip, and Horizontal Flip. This resulted in the generation of approximately 26,514 samples for training in each iteration of 10-fold cross-validation (4419×6). The results for 10-fold cross-validation are reported in Table 9. For each iteration, EfficientNet-B2 was trained for 15 epochs, therefore, it was exhaustive validation and a very computationally expensive approach to opt for.

The evaluation in Table 9 shows that the model was rigorously

validated, however, in each iteration model somehow showed consistent performance.

3.5. Web-Server

For public usage and demo, the web server of EfficientSkinDis was hosted on Streamlit. Here, users can upload images and can get predictions. The interface can be seen in Fig. 13, while the server can be accessed at (<https://abdulrafay97-efficientskindis-app-oncon.streamlit.app/>).

3.6. Discussion

The goal of this study was to develop a classification model for 31 skin diseases to identify and categorize them by analyzing images of the affected areas. The dataset which was curated from Atlas Dermatology had intra-class diversity, as some of the classes contained pictures of the face as well as of other human body parts. The proposed model namely EfficientSkinDis can assist medical professionals in diagnosing skin conditions and creating treatment plans. It can also enhance the diagnostic process by making it faster and more accurate, ultimately providing better care for patients. Furthermore, the model can contribute to research and further understanding of skin diseases by accurately identifying and classifying them.

Table 10

Comparative analysis with previous state-of-the-art skin disease classification studies using deep learning.

Ref	Year	Authors	Dataset	Model	Accuracy Scores
			Manually curated from Atlas Dermatology and ISIC Skin Disease Dataset	Fine-tuned EfficientNet-B2	Top-1 Accuracy: 87.15%
[17]	2022	Karthik et al.	Dataset of (Acne, Actinic Keratosis (AK), Melanoma, and Psoriasis)	EfficientNetV2	84.70%
[41]	2022	Aboulmira et al.	Dermnet	DenseNet	68.97%
[42]	2020	Bajwa et al.	DermNet Database	Fine-tuned DenseNet-161, SE-ResNeXt-101 and NASNet	80%
[43]	2019	Sah et al.	Dermnet Database	Fine-tuned pretrained VGG16 model	76.3%
[44]	2017	Esteve et al.	Clinical images Dataset	Pretrained Inception v3	55.4 ± 1. %
[45]	2016	Liao et al.	AtlasDerm Derma DermIS Dermnet DermQuest	AlexNet model trained from scratch and using fine-tuning	Top-1 Accuracy: 27.6% Top-5 Accuracy: 57.9%
[46]	2016	Kawahara et al.	Dermofit	Pretrained Alex-Net	81.8%
[47]	2015	Liao	Dermnet OLE	Deep convolutional neural network (CNN) using VGG19	Top-1 Accuracy: 73.1% Top-5 Accuracy: 91.0%

In recent years, deep learning has emerged tremendously and various studies have opted for deep neural networks such as in the area of Computer Vision, Biometrics, Biomedical Image analysis and Bioinformatics [35–38]. ResNet [39], VGG [40], and EfficientNet [33] are all convolutional neural network (CNN) architectures designed for image classification. ResNet [39] is a neural network architecture that uses skip connections to address the problem of vanishing gradients in deep neural networks, as shown in Fig. 14. These connections allow gradients to bypass certain layers, enabling the network to learn even very deep architectures. The name “ResNet” comes from the fact that it uses residual connections.

VGG [40] is a convolutional neural network architecture that is widely used as a reference for comparing the performance of other networks. It has a simple structure consisting of a stack of convolutional layers with small filters (3x3), which makes it powerful but computationally expensive due to the large number of parameters it includes, as shown in Fig. 15. The name “VGG” comes from the name of the founding research group, Visual Geometry Group at Oxford.

EfficientNet [33] is a new neural network architecture that aims to improve the performance of existing CNNs while being more efficient in terms of the number of parameters and computational cost. It achieves this by using a combination of scaling and depth-wise convolution, allowing it to achieve good performance with fewer parameters, as shown in Fig. 16. This makes it a good choice for applications that have limited computational resources. It was developed recently and is therefore considered to be a state-of-the-art architecture.

There are over 1500 known skin disorders, which can range from mild to severe and can significantly impact a person’s quality of life [41]. Skin disease diagnosis can be challenging because of the many

visual factors that can influence it, such as the texture of the skin and the presence of hair. This study is a major contribution to the medical industry and as well to the research community. Previously, various studies have contributed to the field of skin disease diagnosis, however, either the proposed models lacked in terms of accuracy or they lacked in terms of the number of skin diseases considered in the dataset. A comparative analysis of existing studies with the present study is shown in Table 10.

The most recent study was from Karthik et al., in which the authors trained a deep convolutional neural network (CNN) model only on 4 diseases which were Acne, Actinic Keratosis, Melanoma, and Psoriasis, however, the model achieved a testing accuracy of 85.70%. The study with most disease classes as per our knowledge was of Aboulmira et al., in 2022, as the model was trained on 23 types of skin diseases using the “Dermnet” dataset. The results showed that the Dense-Net architecture performed the best, with a Top-1 accuracy of 68.97% and a Top-5 accuracy of 89.05% [41]. However, the Top-1 accuracy score was not up to an extent.

In contrast, in the present study i.e., all evaluations were performed based on the Top-1 score and EfficientSkinDis outperformed the existing studies with a higher accuracy score. Also, the number of diseases in the dataset was higher as compared to other studies.

4. Conclusion

Herein, the present study proposed a model namely EfficientSkinDis for the classification of 31 skin diseases. The final model was successful in achieving 87.15% accuracy in identifying 31 skin diseases when evaluated on testing data. For evaluation, four different splitting and augmentation schemes were used. Based on the evaluation, it is concluded that the proposed model can benefit society by enabling doctors to diagnose skin conditions more accurately and quickly. This can lead to faster and more effective treatment, improving patients’ quality of life and reducing the burden on the healthcare system. The model available on the webserver can also benefit the general public by helping them diagnose and gather information about their skin conditions. This can enable them to seek appropriate treatment and to better understand how to prevent and manage these conditions.

As previously stated, that dataset was a major problem faced during this study. It is well known that there are not only 31 skin diseases in the world. A person can experience a lot of different skin conditions. Furthermore, there is still a gap for improvement in terms of the accuracy of the model. Thus, in future, we will add a dataset of additional diseases to the dataset used in this study and train the model on this newly assembled dataset. This will make the model more comprehensive and reliable.

CRedit authorship contribution statement

Abdul Rafay: Methodology, Software, Validation, Investigation, Resources, Data curation. **Waqar Hussain:** Conceptualization, Validation, Formal analysis, Investigation, Resources, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset is available on Google Drive (https://drive.google.com/drive/folders/1AiDVpgy-o4ZLKXZ_yqnWWEHFYhCbrfP1).

References

- [1] W.D. James, D. Elston, T. Berger, Andrew's diseases of the skin E-book: clinical dermatology, Elsevier Health Sciences, 2011.
- [2] S. Chakraborty, K. Mali, S. Chatterjee, S. Anand, A. Basu, S. Banerjee, M. Das, A. Bhattacharya, Image based skin disease detection using hybrid neural network coupled bag-of-features, IEEE 8th Annual Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON), IEEE 2017 (2017) 242–246.
- [3] B. Mondal, N. Das, K. Santosh, M. Nasipuri, Improved Skin Disease Classification Using Generative Adversarial Network, 2020 IEEE 33rd International Symposium on Computer-Based Medical Systems (CBMS), IEEE, 2020, pp. 520–525.
- [4] R. Hay, M. Augustin, C. Griffiths, W. Sterry, B.o.t.I.L.o.D. Societies, t.G.C.C. groups, K. Abuabara, M. Airoidi, F. Ajose, S. Albert, A. Armstrong, The global challenge for skin health, Blackwell Publishing Ltd Oxford, UK, 2015, pp. 1469–1472.
- [5] K. Skowron, J. Bauza-Kaszewska, Z. Kraszewska, N. Wiktorczyk-Kapischke, K. Grudlewska-Buda, J. Kwiecińska-Piróg, E. Walecka-Zacharska, L. Radtke, E. Gospodarek-Komkowska, Human skin microbiome: impact of intrinsic and extrinsic factors on skin microbiota, Microorganisms 9 (2021) 543.
- [6] S. Inthiyaz, B.R. Altahan, S.H. Ahammad, V. Rajesh, R.R. Kalangi, L.K. Smirani, M. A. Hossain, A.N.Z. Rashed, Skin disease detection using deep learning, Adv. Eng. Softw. 175 (2023), 103361.
- [7] H. Xu, H. Li, Acne, the skin microbiome, and antibiotic treatment, Am. J. Clin. Dermatol. 20 (2019) 335–344.
- [8] D.K. Langemo, G. Brown, Skin fails too: acute, chronic, and end-stage skin failure, Adv. Skin Wound Care 19 (2006) 206–212.
- [9] K. Clarysse, M. Grosber, J. Ring, J. Gutermuth, C. Kivlahan, Skin lesions, differential diagnosis and practical approach to potential survivors of torture, J. Eur. Acad. Dermatol. Venereol. 33 (2019) 1232–1240.
- [10] Y. Sawada, N. Saito-Sasaki, E. Mashima, M. Nakamura, Daily Lifestyle and Inflammatory Skin Diseases, Int. J. Mol. Sci. 22 (2021) 5204.
- [11] L. Alzubaidi, J. Zhang, A.J. Humaidi, A. Al-Dujaili, Y. Duan, O. Al-Shamma, J. Santamaria, M.A. Fadhel, M. Al-Amidie, L. Farhan, Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions, J. Big Data 8 (2021) 1–74.
- [12] M.N. Saadat, M. Shuaib, Advancements in deep learning theory and applications: Perspective in 2020 and beyond, Adv. Appl. Deep Learn. 3 (2020).
- [13] G. Litjens, T. Kooi, B.E. Bejnordi, A.A.A. Setio, F. Ciompi, M. Ghafoorian, J.A. Van Der Laak, B. Van Ginneken, C.I. Sánchez, A survey on deep learning in medical image analysis, Med. Image Anal. 42 (2017) 60–88.
- [14] G. Currie, K.E. Hawk, E. Rohren, A. Vial, R. Klein, Machine learning and deep learning in medical imaging: intelligent imaging, J. Med. Imaging Radiat. Sci. 50 (2019) 477–487.
- [15] D. Shen, G. Wu, H.-I. Suk, Deep learning in medical image analysis, Annu. Rev. Biomed. Eng. 19 (2017) 221–248.
- [16] S. Lu, Z. Zhu, J.M. Gorris, S.H. Wang, Y.D. Zhang, NAGNN: classification of COVID-19 based on neighboring aware representation from deep graph neural network, Int. J. Intell. Syst. 37 (2022) 1572–1598.
- [17] R. Karthik, T.S. Vaichole, S.K. Kulkarni, O. Yadav, F. Khan, Eff2Net: An efficient channel attention-based convolutional neural network for skin disease classification, Biomed. Signal Process. Control 73 (2022), 103406.
- [18] V.R. Allugunti, A machine learning model for skin disease classification using convolution neural network, Int. J. Comput. Programm. Database Manage. 3 (2022) 141–147.
- [19] M.N. Hossen, V. Panneerselvam, D. Koundal, K. Ahmed, F.M. Bui, S.M. Ibrahim, Federated machine learning for detection of skin diseases and enhancement of internet of medical things (IoMT) security, IEEE J. Biomed. Health Informatics (2022).
- [20] G. Cai, Y. Zhu, Y. Wu, X. Jiang, J. Ye, D. Yang, A multimodal transformer to fuse images and metadata for skin disease classification, Vis. Comput. (2022) 1–13.
- [21] V. Anand, S. Gupta, D. Koundal, S.R. Nayak, J. Nayak, S. Vimal, Multi-class Skin Disease Classification Using Transfer Learning Model, Int. J. Artif. Intell. Tools 31 (2022) 2250029.
- [22] P.R. Kshirsagar, H. Manoharan, S. Shitharth, A.M. Alshareef, N. Albishry, P. K. Balachandran, Deep Learning Approaches for Prognosis of Automated Skin Disease, Life 12 (2022) 426.
- [23] S.-Y. Lu, Z. Zhang, Y.-D. Zhang, S.-H. Wang, CGENet: A deep graph model for COVID-19 detection based on chest CT, Biology 11 (2022) 33.
- [24] P.N. Srinivasu, J.G. SivaSai, M.F. Ijaz, A.K. Bhoi, W. Kim, J.J. Kang, Classification of skin disease using deep learning neural networks with MobileNet V2 and LSTM, Sensors 21 (2021) 2852.
- [25] S. Lu, S.-H. Wang, Y.-D. Zhang, Detection of abnormal brain in MRI via improved AlexNet and ELM optimized by chaotic bat algorithm, Neural Comput. & Applic. 33 (2021) 10799–10811.
- [26] S. Voggu, K.S. Rao, A Survey on Skin Disease Detection Using Deep Learning Techniques, J. Algebraic Statistics 13 (2022) 3916–3920.
- [27] P.M. Burlina, N.J. Joshi, E. Ng, S.D. Billings, A.W. Rebman, J.N. Aucott, Automated detection of erythema migrans and other confounding skin lesions via deep learning, Comput. Biol. Med. 105 (2019) 151–156.
- [28] M. Monisha, A. Suresh, M. Rashmi, Artificial intelligence based skin classification using GMM, J. Med. Syst. 43 (2019) 1–8.
- [29] R. Bhavani, V. Prakash, R. Kumaresh, R. Sundra Srinivasan, Vision-based skin disease identification using deep learning, Int. J. Eng. Adv. Technol. (IJEAT) 8 (2019) 2249–8958.
- [30] N.S.A. ALEnezi, A method of skin disease detection using image processing and machine learning, Procedia Comput. Sci., 163 (2019) 85–92.
- [31] J. Velasco, C. Pascion, J.W. Alberio, J. Apuang, J.S. Cruz, M.A. Gomez, B. Molina Jr, L. Tuala, A. Thio-ac, R. Jorda Jr, A smartphone-based skin disease classification using mobilenet cnn, arXiv preprint arXiv:1911.07929, (2019).
- [32] N. Hameed, A.M. Shabut, M.A. Hossain, Multi-class skin diseases classification using deep convolutional neural network and support vector machine, 2018 12th International Conference on Software, Knowledge, Information Management & Applications (SKIMA), IEEE, 2018, pp. 1–7.
- [33] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, International conference on machine learning, PMLR, 2019, pp. 6105–6114.
- [34] K. Ali, Z.A. Shaikh, A.A. Khan, A.A. Laghari, Multiclass skin cancer classification using EfficientNets—a first step towards preventing skin cancer, Neuroscience Informatics 2 (2022), 100034.
- [35] W. Hussain, sAMP-PFPDeep, Improving accuracy of short antimicrobial peptides prediction using three different sequence encodings and deep neural networks, Brief. Bioinform. 23 (2022) bbab487.
- [36] W. Hussain, N. Rasool, M. Yaseen, ADVIT: Using the potentials of deep representations incorporated with grid-based features of dorsum vein patterns for human identification, Forensic Sci. Int. 313 (2020), 110345.
- [37] M. Shahid, M. Ilyas, W. Hussain, Y.D. Khan, ORI-Deep: improving the accuracy for predicting origin of replication sites by using a blend of features and long short-term memory network, Brief. Bioinform. 23 (2022) bbac001.
- [38] U. Hasan, W. Hussain, N. Rasool, AEPI: insights into the potential of deep representations for human identification through outer ear images, Multimed. Tools Appl. 81 (2022) 10427–10443.
- [39] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 770–778.
- [40] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint arXiv:1409.1556, (2014).
- [41] A. Aboulmira, H. Hrimech, M. Lachgar, Comparative Study of Multiple CNN Models for Classification of 23 Skin Diseases, Int. J. Online Biomed. Eng. 18 (2022).
- [42] M.N. Bajwa, K. Muta, M.I. Malik, S.A. Siddiqui, S.A. Braun, B. Homey, A. Dengel, S. Ahmed, Computer-aided diagnosis of skin diseases using deep neural networks, Appl. Sci. 10 (2020) 2488.
- [43] A.K. Sah, S. Bhusal, S. Amatya, M. Mainali, S. Shakya, Dermatological diseases classification using image processing and deep neural network, in: 2019 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS), 2019, pp. 381–386.
- [44] A. Esteve, B. Kuprel, R.A. Novoa, J. Ko, S.M. Swetter, H.M. Blau, S. Thrun, Dermatologist-level classification of skin cancer with deep neural networks, Nature 542 (2017) 115–118.
- [45] H. Liao, Y. Li, J. Luo, Skin disease classification versus skin lesion characterization: Achieving robust diagnosis using multi-label deep neural networks, in: 2016 23rd International Conference on Pattern Recognition (ICPR), 2016, pp. 355–360.
- [46] J. Kawahara, A. BenTaieb, G. Hamarneh, Deep features to classify skin lesions, IEEE 13th international symposium on biomedical imaging (ISBI), IEEE 2016 (2016) 1397–1400.
- [47] H. Liao, A deep learning approach to universal skin disease classification, University of Rochester Department of Computer Science, CSC, 2016.